

Ethan Shen

ethans03@cs.washington.edu · ethanlshen.github.io · [Google Scholar](#) · [GitHub](#)

Research Interests

I am currently working on synthetic program creation, on-policy distillation, and studying the importance of verification during post-training. Broadly, my research interests are at the intersection of coding agents [1,2], training dynamics [3,4,5], and data curation [1,2]. I am also interested in learning more about systems engineering and AI policy.

Education

University of Washington

Sep 2025 – Present (Expected Graduation: Summer 2029)

Ph.D. Candidate in Computer Science · Advisor: Ali Farhadi

– Fellow, Paul and Daisy Soros Fellowship for New Americans

– Awardee, NSF Graduate Research Fellowship Program

University of Washington

Aug 2022 – Jun 2025

Sc.B. in Computer Science (3.96 GPA)

Research and Work Experience

Research Intern, Allen Institute for AI (AI2)

Feb 2025 – Present

Leading development of Ai2's coding agents starting with SERA [2], a SoTA open-source coding agent with 8B/32B parameters. Mentored by Tim Dettmers.

Software Engineer Intern, Amazon

Jun 2024 – Aug 2024

Built LLM pipeline with AWS Bedrock, Lambda, and DynamoDB to automate invoice processing, saving 1200+ hours of manual work per year.

Machine Learning Intern, Papyrus

Dec 2023 – Mar 2024

Worked at pre-seed startup and created primary ASR pipeline for speaker labeling, transcription, and diarization using Whisper, Demucs, and Pyannote.

Undergraduate Research Assistant, RAIVN Lab – UW

Jun 2023 – Aug 2025

Led and assisted multiple projects, such as the first application of latent thinking in LLMs [4]; task-specific tokens for VLM reasoning [3]; and investigated tradeoffs of different geometric representations inside models [5]. Mentored by Aditya Kusupati and Matt Wallingford.

Selected Publications

[1] [OpenThoughts-Agent: Data Recipes for Agentic Models](#)

N. Raoof*, R. Zhuang*, M. Nezhurina*, ..., **Ethan Shen**, ..., A. Dimakis, B. Feuer, L. Schmidt · *Preprint 2026*

A fully open data-curation pipeline and systematic recipe study for training broadly capable agentic models across diverse tasks.

[2] **SERA: Soft-Verified Efficient Repository Agents**

Ethan Shen, D. Tormoen, S. Shah, A. Farhadi, T. Dettmers · *ICML 2026*

An efficient synthetic-data method that uses soft verification to train and specialize repository agents without requiring unit tests, producing SoTA specialized and general coding agents.

[3] **Perception Tokens Enhance Visual Reasoning in Multimodal Language Models**

M. Bigverdi, Z. Luo, C.-Y. Hsieh, **Ethan Shen**, D. Chen, L. Shapiro, R. Krishna · *CVPR 2025*

A framework that equips multimodal language models with learned depth and bounding-box tokens as intermediate steps for visual reasoning.

[4] **Superposed Decoding: Multiple Generations from a Single Autoregressive Inference Pass**

Ethan Shen, A. Fan, S. Pratt, J. S. Park, M. Wallingford, S. Kakade, A. Holtzman, R. Krishna, A. Farhadi, A. Kusupati · *NeurIPS 2024*

A decoding algorithm that generates k drafts at the cost of a single autoregressive inference pass via superposition of token embeddings.

[5] **Are “Hierarchical” Visual Representations Hierarchical?**

Ethan Shen, A. Farhadi, A. Kusupati · *Workshop on Symmetry and Geometry in Neural Representations @ NeurIPS 2023*

An empirical benchmark showing that explicitly hierarchical visual representations do not capture human-perceived visual hierarchies better than standard representations.

Invited Talks

SERA: Soft-Verified Efficient Repository Agents

– Bosch Center for Artificial Intelligence (Host: Bingqing Chen)

Superposed Decoding

- AWS AI Labs CodeGen (Host: Zijian Wang)
- Amazon LLM Reasoning Group (Host: Linbo Liu)
- AWS AI Labs Quicksight (Host: Patrick Ng)

Service

Reviewing

ICML — 2026 (Silver Reviewer)

NeurIPS — 2025, 2026

EMNLP — 2024, 2025

Teaching

CSE 493: Deep Learning — TA (Winter 2026)

CSE 190B: CSE Freshman Seminar — TA (Autumn 2023)

References

Ali Farhadi

Professor, University of Washington

Corporate Vice President, Microsoft AI

Tim Dettmers

Assistant Professor, Carnegie Mellon University

Research Scientist, Allen Institute of AI

Aditya Kusupati

Staff Research Scientist, Google Deepmind